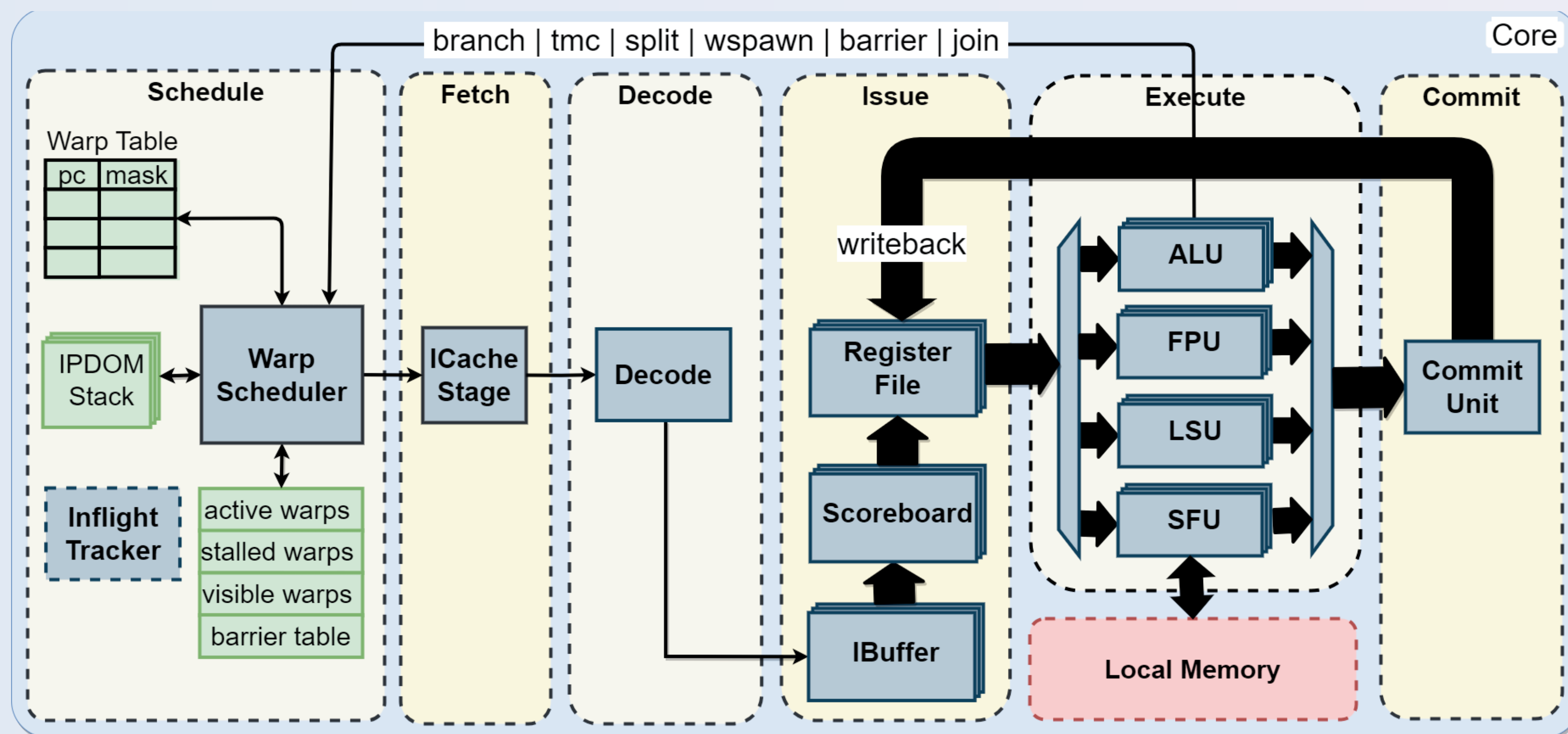




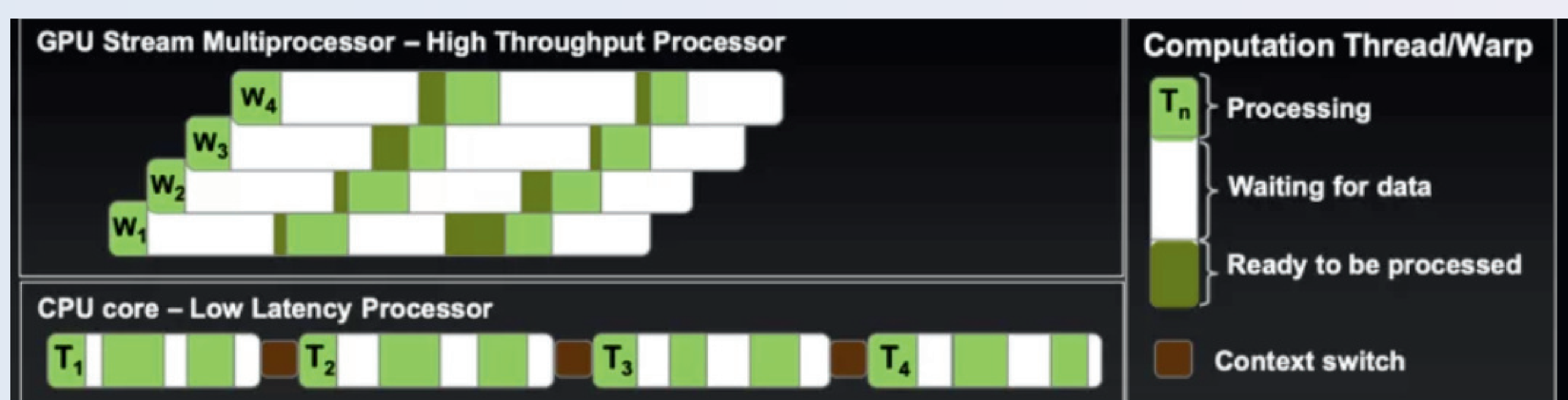
ASIC Design and Performance Optimization of Vortex (Open-Source RISC-V GPGPU)

Graduation Project Year 2025/2026
Supervised by: Dr. DiaaEldin Khalil & Hossam Hassan
Sponsored by : silicon Arts

Introduction



- In this project, we implement an Easy Flow execution model, derived from the Vortex architecture, with a focus on simplicity, clarity, and educational value.
- The proposed architecture follows a SIMT (Single Instruction, Multiple Threads) execution model, where groups of threads, organized as warps, execute the same instruction stream in lockstep while maintaining independent execution states.
- Overview of RISC-V GPU Architecture & Difference Between RISC-V GPU and CPU Architectures :



Methodology

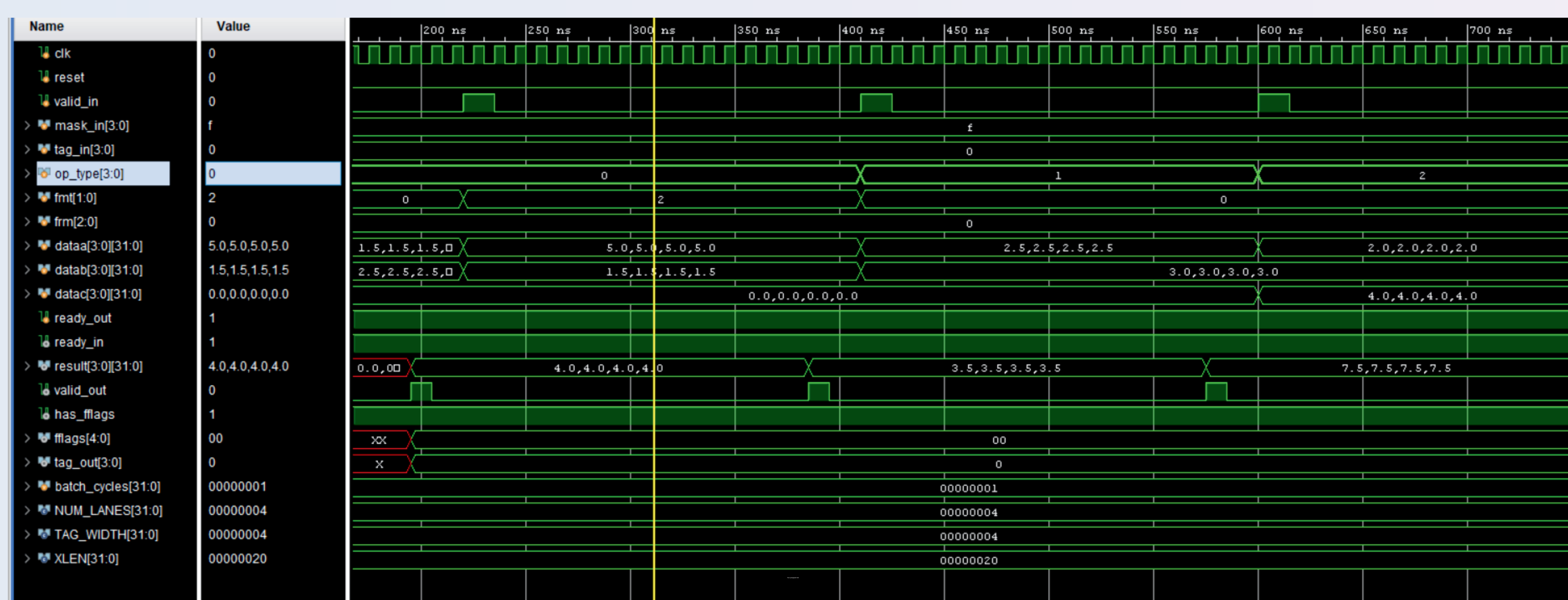
- The proposed RISC-GPU is implemented using Verilog and SystemVerilog RTL based on the Vortex architecture.
- Synopsys IC Compiler (ICC)
- Vivado Xilinx on two modules before using tools provided by DR Hossam Hassan
- Functional verification is performed through simulation using EDA tools and Questasim

Future Work

- Complete Full-Chip Synthesis
- Verification of the modified codes
- Baseline Physical Implementation
- Microarchitectural Optimizations
- Signoff & Documentation

Results

- We successfully synthesized and verified the Vortex execution unit using a comprehensive testbench environment integrated with industry-standard ASIC tools (Synopsys DC/PrimeTime flow).
- The Results :



- These results highlight the effectiveness of the Easy Flow approach in maintaining functional correctness while simplifying GPU control logic

Conclusion

- The project validated the Vortex execution unit through a rigorous industry-standard ASIC flow (Synopsys DC/PrimeTime), achieving functional correctness and timing closure.

Presented by :

Mark Maher 21P0355 , Eiz Eldeen Housam 21P0170
Sama Ashraf 21P0066 , Abdelhmed Mahmoud 21P0171
Amir Raed 21P0380 ,Youssef Khaled 22P0278
Ahmed Mostafa 21P0407 , Abdullah Ahmed 2100369